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**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
**Off-Campus Centre of Nitte (Deemed to be University)**  
**Fourth Semester B.Tech (CBCS) Degree Examinations**

May 2025

IS1103-1 – INTRODUCTION TO SOFTWARE ENGINEERING

(For IS)

Duration: 3 Hours

Max. Marks: 100

**Note:**

**Part – A: Multiple Choice Questions:** Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

**Part – B: Descriptive Answer Questions:** Answer **Five full** questions, choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

**PART - A: MULTIPLE CHOICE QUESTIONS****20 Marks**

1. Which of the following is NOT an essential attribute of good software?  
A) Maintainability B) Security  
C) Inefficiency D) Acceptability
2. Which of the following is NOT a primary software process activity?  
A) Software specification B) Software development  
C) Software marketing D) Software validation
3. What is a key challenge in software engineering?  
A) Finding software engineers B) Ensuring software reliability and security  
C) Writing software without documentation D) Using as many programming languages as possible
4. Which factor affects most modern software development?  
A) Heterogeneity B) Only using one programming language  
C) Avoiding client feedback D) Ignoring security concerns
5. What is the main purpose of system modeling in software engineering?  
A) To test software performance B) To represent the structure and behavior of a system  
C) To document source code D) To debug software
6. What does architectural design in software engineering focus on?  
A) The physical structure of a system B) The high-level organization of software components  
C) The detailed coding structure D) The debugging process
7. What is the key difference between Agile and Plan-Driven Development?  
A) Agile focuses on flexibility, while Plan-Driven follows a fixed structure B) Plan-Driven development does not involve any documentation  
C) Agile never uses project planning D) Plan-Driven development does not require team collaboration
8. Which of the following is an Agile methodology?  
A) Scrum B) Waterfall  
C) V-Model D) Spiral Model
9. Which UML diagram is used to model object interactions over time?  
A) Class Diagram B) Sequence Diagram  
C) Deployment Diagram D) Activity Diagram

10. Which is NOT a requirements elicitation technique?
- A) Interviews B) Surveys  
C) Coding D) Prototyping
11. Which of the following is NOT a reason for incomplete requirements?
- A) Lack of stakeholder involvement B) Changes in business needs  
C) Poor software testing D) Miscommunication among teams
12. What is the main advantage of the layered architecture pattern?
- A) Reduces software development time B) Enhances code reusability and maintainability  
C) Eliminates the need for databases D) Avoids user interaction
13. Which of the following is the most important factor when managing people in a project?
- A) Allocating work equally among team members B) Maintaining open communication  
C) Offering financial incentives D) Focusing solely on the technical skills of the team
14. What is the primary purpose of project scheduling in project management?
- A) To create a budget B) To define project risks  
C) To organize tasks and allocate resources D) To outline project objectives
15. What is the focus of "Managing People" in project management?
- A) Ensuring that the team follows a strict schedule B) Motivating and guiding team members to achieve project goals  
C) Monitoring financial expenses D) Developing technical skills only
16. What does the term "Agile methodology" emphasize in project management?
- A) Rigid schedules and detailed documentation B) Flexibility and customer collaboration  
C) Predictability and predefined outcomes D) Fixed scope and goals
17. What does "resource leveling" refer to in project scheduling?
- A) The process of adjusting the project schedule to balance resource allocation B) The process of reducing project scope  
C) The process of defining roles and responsibilities D) The process of reducing project costs
18. Which of the following is the first step in risk management?
- A) Risk monitoring B) Risk mitigation  
C) Risk identification D) Risk documentation
19. What is the role of a project sponsor in project management?
- A) To manage day-to-day project tasks B) To provide financial and strategic support for the project  
C) To create detailed project schedules D) To perform risk management tasks
20. What is the purpose of a Gantt chart in project scheduling?
- A) To track project costs B) To identify risks and issues  
C) To visually represent the project schedule and tasks D) To monitor team performance

**PART - B: DESCRIPTIVE ANSWER QUESTIONS**

| <b>Unit – I</b>   |                                                                                                                                                                                                                                                                                                              | <b>Marks</b> | <b>BT*</b> | <b>CO*</b> | <b>PO*</b> |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------|------------|------------|
| 1.                | a) Explain Incremental development, its advantages and disadvantages.                                                                                                                                                                                                                                        | 08           | L*2        | 2          | 1          |
|                   | b) Explain the role of Requirements Engineering in software development. Discuss the different types of requirements with suitable examples. Also, describe the key challenges faced during the requirements engineering process.                                                                            | 08           | L2         | 1          | 1          |
| 2.                | a) Explain the requirements collection process through Interviews, highlighting their advantages and challenges.                                                                                                                                                                                             | 08           | L2         | 1          | 1          |
|                   | b) Explain the phases of the Waterfall Model with a well-labeled diagram. Critically evaluate its applicability in large-scale and high-risk projects. Additionally, propose modifications or hybrid approaches that can overcome its drawbacks.                                                             | 08           | L3         | 2          | 2          |
| 3.                | a) The Software Requirements Specification (SRS) document plays a crucial role in software engineering by ensuring that all stakeholders have a shared understanding of system requirements.                                                                                                                 |              |            |            |            |
|                   | (i) Explain the key components of an SRS document, emphasizing their significance in software development.                                                                                                                                                                                                   |              |            |            |            |
|                   | (ii) Discuss the importance of Requirements Specification and highlight two common challenges faced during the specification process.                                                                                                                                                                        | 08           | L3         | 1          | 2          |
|                   | b) Explain the types of nonfunctional requirements.                                                                                                                                                                                                                                                          | 08           | L2         | 1          | 1          |
| <b>Unit – II</b>  |                                                                                                                                                                                                                                                                                                              |              |            |            |            |
| 4.                | a) Consider a software system for an online banking application. Develop a structural model using UML class diagrams and explain how different classes interact. Identify at least four key classes and their relationships. Also, justify how this structural model helps in system design and maintenance. | 08           | L3         | 3          | 2          |
|                   | b) Explain Layered architecture and its advantages and disadvantages.                                                                                                                                                                                                                                        | 08           | L2         | 3          | 1          |
| 5.                | a) Consider an online food delivery system. Develop an event-driven model for the system using state diagrams. Identify key events and states involved in order processing.                                                                                                                                  | 08           | L3         | 3          | 2          |
|                   | b) Explain the scrum process with a neat diagram. Briefly explain the characteristics of the scrum process.                                                                                                                                                                                                  | 08           | L2         | 4          | 1          |
| 6.                | a) Explain Extreme programming. Describe at least 4 principles of the same.                                                                                                                                                                                                                                  | 08           | L2         | 4          | 1          |
|                   | b) Explain Transaction processing systems w.r.t banking application.                                                                                                                                                                                                                                         | 08           | L2         | 3          | 1          |
| <b>Unit – III</b> |                                                                                                                                                                                                                                                                                                              |              |            |            |            |
| 7.                | a) Explain the categories of risks. Consider a Scenario: A multinational company plans to expand its operations into a new country. As a risk manager, identify and explain the key risks involved in this expansion.                                                                                        | 08           | L3         | 5          | 2          |
|                   | b) Explain the project scheduling process with a diagram                                                                                                                                                                                                                                                     | 08           | L2         | 5          | 1          |
| 8.                | a) Explain the factors affecting the efficiency of the group communication                                                                                                                                                                                                                                   | 08           | L2         | 5          | 1          |
|                   | b) Explain algorithmic cost modeling to predict project costs.                                                                                                                                                                                                                                               | 08           | L2         | 5          | 1          |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome.

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